Institutional Entrepreneurship, Wikipedia, and the Opportunity of the Commons*

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Abstract

Wikipedia, the free online encyclopedia, stands as a highly visible success story of an organization that provides a public good, when formal theory implies it should not exist. Rather than focusing on incentivizing expression and minimizing free rider problems through copyright law, some entrepreneurial changes to institutions and rules allow for spaces that can provide expression while minimizing the associated problems – Wikipedia is one such example. I offer an institutional approach, following the Bloomington School's IAD framework, to outline principles of institutions which can successfully manage a commons of expressive works.

JEL Classification: H41, K29
Keywords: Wikipedia, intellectual property rights, copyright, public goods, institutions, commons

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1 Introduction

Imagine it is 1995, and someone proposes to construct an encyclopedia on the internet that is free for everyone to read, has very weak filters, few editors, no peer review, and is open to expert and uncertified amateur alike to publish articles anonymously on whatever they wanted, from the Frisch elasticity of labor supply, to the Loch Ness Monster, to a list of animals with fraudulent diplomas.[1] It would have barely passed the giggle test.

Economists would worry about the "public good" aspects about original articles (Arrow, 1962; Nordhaus, 1969; Besen and Raskind, 1991). An encyclopedia would be one large collective action problem, where the benefits of reading the encyclopedia (for free) are widely dispersed among anonymous readers and the costs of putting in the (uncompensated) time and effort of researching and writing articles is concentrated on individual writers (Olson, 1965). Expressive works like original encyclopedia articles feature high fixed costs of production and low marginal costs of distribution and use. It requires significant effort and investment of time, research, experimentation, and capital to produce an expressive work, but as expressive works are nonrival in consumption – that is, one person's reading of an article does not prevent others from reading the article simultaneously, especially if it is online – the original producer must worry about "free riders." If enough users are able to consume the work without paying (via copying, sharing, or otherwise consuming without payment), then the original author is unable to recoup her capital costs. With an expected negative return, she would have little incentive to produce in the first place. Copyright laws are intended to prevent this from happening by coercively encouraging free riders to purchase the good rather than suffer legal sanctions. In practice, such laws establish a tradeoff between generating the incentive to produce original works by legally prohibiting free riders from unauthorized uses and subsequently reducing access to existing works as a result of the copyright holder's veto power (Landes and Posner, 1989).

Furthermore, anyone would recognize that without strong, centralized control, abuse is likely, if not guaranteed, to be flagrant. Users would pass rumor and fancy as fact, and the careful, dispassionate analysis of scholarly experts would be buried by the demagogues, the quacks, and the mob. Not to mention, without a central manager, negative network externalities are likely to emerge, where different users would create their own works according to different standards and levels of quality, reducing consistency and potentially interoperability with other works or systems.

Perhaps as a response to these perfectly rational and legitimate fears, Jimmy Wales launched Nupedia, "the open content encyclopedia," online in March of 2000.[2] While Nupedia was written by volunteers, and

[1] For the hypothetical, see Boyle (2007, 128ff), Benkler (2002); Litman (2004).

[2] See the memoir from its editor, Larry Sanger, at `http://features.slashdot.org/story/05/04/18/164213/the-early-history-of-nupedia-and-wikipedia-a-memoir`.

ultimately licensed as free content to readers, the content was reviewed by expert editors prior to official publication, and the site expected that contributors would be Ph.Ds or experts in their respective fields. Under editor-in-chief Larry Sanger, Nupedia implemented a seven step peer review process that gradually filtered out low-quality articles. In its first year, Nupedia approved publication of just 21 articles.

As a response to Nupedia's slow productivity, in early 2001, Wales and Sanger founded Wikipedia as an alternative to further open up the flow of articles. In contrast to the centralized management of Nupedia, Wikipedia utilized a new, more open-ended and decentralized web application known as the "wiki." This system allows multiple users to edit a single article without it having a pre-defined structure or owner, archives each change, and allows for users to discuss and reverse changes. In its first month alone, Wikipedia published 200 articles, and reached 18,000 in its first year.

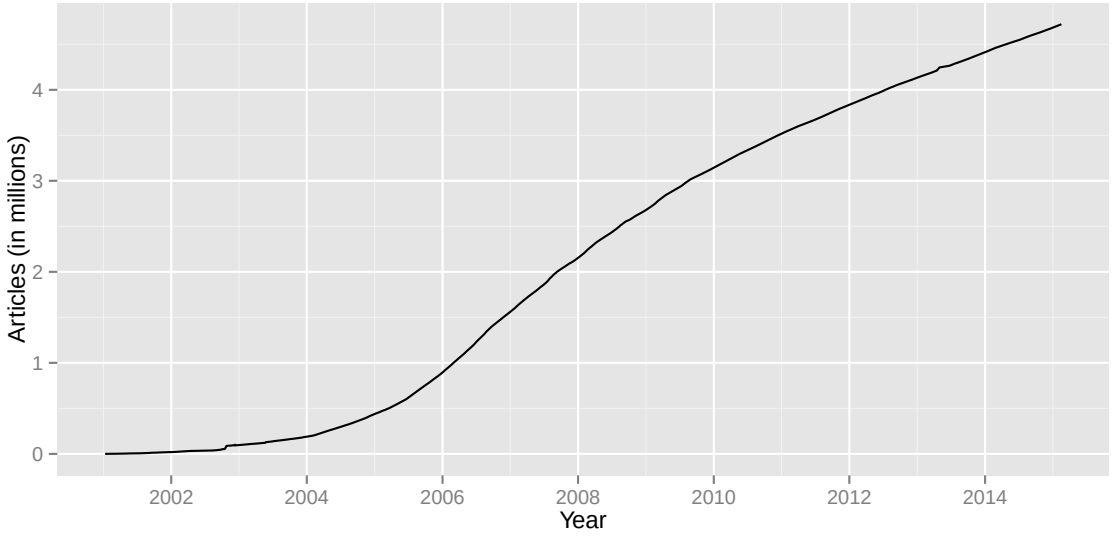


Figure 1: Number of articles on Wikipedia since its creation

As of this writing, Wikipedia is the seventh most visited website in the world.[3] By its own count (as of January 27, 2015), it has 36,011,921 articles in 289 languages (4,965,272 of them in English, the largest Wikipedia) and 57,393,797 users making 2,061,386,730 edits.[4] The leading encyclopedia in the English-speaking world, the famed *Encyclopædia Britannica*, on the other hand, had at its final printing (in 2010), 40,000 articles, and about 120,000 online. Figure 1 shows stunning success, compiled from Wikipedia's own statistics.[5]

The success of Wikipedia and other projects that feature mass-collaboration, open-source, and free soft-

[3] http://www.alexa.com/siteinfo/wikipedia.org
[4] http://meta.wikimedia.org/wiki/List_of_Wikipedias
[5] See http://en.wikipedia.org/wiki/Wikipedia:Size_of_Wikipedia#The_data_set

ware – everything from Linux to R to Mozilla Firefox[6], have lead to a dearth of popular books by scholars, entrepreneurs, and pundits making bold claims about a "new economics" of the internet age (see e.g. Kelly (1999); Raymond (1999); Benkler (2006); Tapscott and Williams (2007); Shirky (2008); Anderson (2008); Downes (2009); Anderson (2009,0)). In any event, the rise of Wikipedia stands at least as an existence proof for the possibility of alternative methods of providing expressive works without strictly following the formal model of intellectual property (IP) rights. Standard theory precludes the existence of Wikipedia due to an inherent free rider problem in the production of expressive works. Without recourse to exclude others from consuming expressive works (through legal fiat, as with copyright laws), producers have no incentive to produce the works in the first place. In fact, what is most interesting about Wikipedia is that it actually exploits copyright law in the *opposite* way the law was intended. Rather than using copyright's legal right to exclude others' ability to consume content to incentivize its original creation, Wikipedia inverts this logic by making all content available for free. Furthermore, it even leverages copyright law to *compel* all of its content, and derivative uses of that content, to be free and nonexcludable. Wikipedia's use of the "copyleft" licenses, under modern copyright law, include stipulations that all content cannot be closed off by any person.[7]

Far too much research in the economics of innovation, art, and intellectual property focuses exclusively on the tradeoff between granting monopoly privileges from intellectual property rights to spur innovation versus encouraging access to existing works (needed to produce derivative works) (Nordhaus, 1969; Besen and Raskind, 1991; Landes and Posner, 1989). The theoretical literature on intellectual property rights in general has focused on determining just what the optimum is (Gilbert and Shapiro, 1990; Klemperer, 1990), but cannot empirical studies have not yielded much insight in such a causally-muddled area (Tschmuck, 2010). This overarching framework is fundamentally incomplete, as it leaves no room for various forms of entrepreneurship which can and does overcome these theoretical problems under a wide range of institutional settings. The existence of thriving legal business models despite strong copyright laws, imperfect enforcement, and extremely low costs of copying are sufficient to demonstrate this fact. Streaming services such as Spotify, Hulu, and Netflix have likely had a strong role in reducing the plague of piracy in music and films (Shen, 2013). Alternatively, as the costs of legally accessing works has increased as copyright law has strengthened in the past 40 years, various methods of accessing works through illegal means (filesharing and torrenting) have also become relatively attractive. Furthermore, many authors give reasons why, even in total absence of copyright law, the number of expressive works might be less than today, but certainly not zero (see e.g.

[6]For a fairly comprehensive list, see `http://en.wikipedia.org/wiki/List_of_free_and_open-source_software_packages`

[7]These licenses are primarily the GNU General Public License(`http://www.gnu.org/copyleft/gpl.html`) and the Creative Commons licenses (`https://creativecommons.org/licenses/`), granting all users the right to use, modify, and distribute existing licensed software and furthermore compel future creators to follow the original license conditions and openly reveal their own source code (see e.g. Benkler, 2002; Boyle, 2008, Ch.7). Under such a license, erecting barriers to distribute, modify, or view the source code violates the compulsory license, and thus is ironically a case of copyright infringement.

Landes and Posner (2003, 41-50); Boldrin and Levine (2008)).

As such, Wikipedia not only stands as a counterexample to the incentive-based logic of intellectual property rights, but a carefully constructed entrepreneurial workaround to the existing system of expressive works production. While in some sense, Wikipedia is a simple case of market entrepreneurship, with Jimmy Wales selling a zero-price encyclopedia, it also exhibits entrepreneurship over "the rules" or the institutional arrangements governing how common resources are provided. There is a growing literature on such "higher-tiered" or "institutional" entrepreneurship, illustrating the importance of examining the choice of alternative rules, as well as strategic choices made under a specific set of rules (Buchanan and Tullock, 1962). All human beings seek ways to maximize their own well-being, and depending on the institutions within a society, such an entrepreneurial drive can manifest itself in productive entrepreneurship on the market, unproductive lobbying for government favors, or through criminal activities that outright destroy wealth (Baumol, 1990). Entrepreneurs can furthermore alter these institutions themselves in order to create a more profitable environment to operate within (Schotter, 1981), by means of establishing new categories of property rights (Alchian, 1965; Demsetz, 1969), removing inefficiencies by creating property rights-protecting technologies in an area lacking state capacity (Li et al., 2006; Leeson and Boettke, 2009), or by changing political rules to reach more favorable political outcomes (Martin and Thomas, 2013). In the context of public goods, Wagner (2013) astutely observes, the problem of free riding is not inherent in the ontology of the good itself, but "an artifact of a particular set of institutional assumptions." Applying this idea to Hardin's original tragedy of the commons example, Cole et al. (2014, 353) similarly highlight that the tragedy results only with a particular set of flawed institutions. Thus, crafting institutions properly to avoid such problems is of prime importance.

What Wikipedia does is create an effective governance mechanism over a common pool resource – free encyclopedia articles. This paper, following Hess and Ostrom (2007b) and Dourado and Tabarrok (2014), draws upon the institutional analysis of Ostrom (1990) and other scholars collectively known as the "Bloomington School" to understand how proper design of informal rules lead communities to manage common pool resources effectively. This approach elucidates eight principles for successful and long-enduring institutions, which I apply to examine the how Wikipedia fits these eight principles, as well as exploring some of the challenges that Wikipedia, both as an organization and as a ruleset for online social interaction, grapples with. I conclude that Wikipedia offers a case study that deserves scholarly attention, and I draw implications for how this should affect our understanding of intellectual property and how to best "promote the Progress of Science and useful Arts" (United States Constitution, Article I, §8).

2 Expressions and the Commons

Many scholars have productively suggested that we can best view culture, expression, and knowledge as a commons (Benkler, 2002; Hess and Ostrom, 2007b; Dourado and Tabarrok, 2014; Schweik and English, 2012; Frischmann et al., 2014).[8] Economists typically define a commons, or a common pool resource, as any resource that is rivalrous but non-excludable – that is, while each person's use of a resource crowds out others' ability to use it, nobody has any ability to prevent others from using such a resource. Hardin (1968) famously describes the potential "tragedy" of the commons whereby, due to the inability to exclude others, each person faces an individual incentive to over-exploit the resource before everyone else does. This ultimately ends up depleting the resource, with common examples being overgrazed pastures, over-fished waters, and polluted environments. The classic solution proffered to correct this solution and internalize these negative externalities is to establish property rights over the resource, granting the ability of each user to exclude other users (Alchian, 1965; Demsetz, 1964,9,9).

On the other hand, recent research in economic development and property rights have highlighted another potential risk of improperly prescribed property rights by governments. If rights are poorly designed and allocated from above, or they are thrown into a hostile institutional framework that has little historical respect for liberty of contract, freedom of association, or judicial enforcement (Boettke et al., 2004), then what results is the symmetrically opposite "tragedy of the *anti*-commons," (Heller, 1997; Buchanan and Yoon, 2000; Heller, 2008). In such a situation, multiple users are granted rights to exclude others from using a resource, requiring a would-be user to secure the blessing of every potential veto-holder that would block her use. The tragic result is that the valuable resource lies dormant, mummified in bureaucratic red tape and holdout problems.

Rather than a strict binary public-vs-private or propertized-vs-nonpropertized distinction that dictates success or failure, the Bloomington School has shown that there is a wide variety of institutional forms operating in between markets and states that can govern a commons (Ostrom, 2010). Indeed, rather than a guaranteed "tragedy," the commons present an "opportunity," provided the affected group can establish effective governance rules over the resource.

In terms of culture, Hess and Ostrom (2007b, 3) apply a rather wide interpretation of a commons, defining it merely as "a resource shared by a group of people that is subject to social dilemmas." This definition allows us to view content such as expressive works as a commons, as the resource of "expressive works," is shared by just about every member of society who values expression, and further by those who benefit from the fruits of science and innovation. The creation of knowledge and expressive works requires access to the

[8] I thank Mike Madison for directing me to recent literature on this.

commons of previous works in order to build upon.

This approach is also amiable as it sidesteps potent criticisms about viewing ethereal expressions and inventions as objects capable of being as legitimately propertized as can a house or a car. Indeed, this might be a misappropriation of a *legal* concept for the *political* benefit of a particular industry (Lessig, 2004; Kinsella, 2008). The rhetoric of property rights can often be a facade for political action used for private gain (Stigler, 1971). Whether for taxi medallion auctions, defense contract procurement, or tradeable pollution permits, the language of property rights and markets often merely consecrates that which is to be perceived as wise and efficient political program, when in fact it is a rent-seeking victory by some interest group. Such opponents prefer the term intellectual *privilege*, both analytically and normatively, to describe the actual dynamic at work (Bell, 2014). Intellectual "property rights" appear in this view to be merely enforceable claims to continue existing business models and veto disruptive innovations (Acemoglu and Robinson, 2000), akin to horse-and-buggy producers having the "property right" to prohibit the automobile from being manufactured. As Landes and Posner (2003, 23) remind us, "[a] fundamental principle of American law is that competition is not a tort, that is, an invasion of a legally protected right." The pioneering work of Arnold Plant (1934, 31) further clarifies that so-called intellectual property rights are peculiar because unlike ordinary property rights in physical goods, IP rights are not a *consequence* of scarcity, they are a deliberate *creation* of scarcity by a legislature.

Rather than traditional fears of overuse and nonsubtractability in resource commons, the knowledge commons can be, in fact, *additive*, or, in the words of Dourado and Tabarrok (2014, 19), "Super-Lockean" as, contingent upon it being relatively accessible, the more people draw from it and share, the more value they add to society (see also (Hess and Ostrom, 2007b, 5)). Knowledge and expressive works, being nonrivalrous, can add more than "enough and as good left" when used, as consuming, replicating, and constructiving derivate works more widely distributes the original. Furthermore, the production of new works requires use of older works as inputs. As Matt Ridley (2011, 7) memorably relates, "ideas have sex with each other," that is, they are combinatorial, and it is only through the exchange of ideas, rather than a single orphan idea from a lone genius, that great technologies are born. The camera pill, for instance, was born out of a conversation between a gastroenterologist and a guided missile designer (Ridley, 2011, 270). Henry Ford is supposed to have said that he "simply assembled into a car the discoveries of other men behind whom were centuries of work." Rather than reinventing the wheel at every iteration, all expressive works require some amount of existing expressive works as inputs, hence the important *access* part of the incentive-access tradeoff. One of the primary challenges of the knowledge commons is to prevent it from becoming an *anti*-commons from an excessive amount of veto points.

As described above, traditional views of commons imply that unpropertized commons are automatically

subject to the tragedy of the commons, where the resource gets depleted due to incentives to over-exploit private uses without concern for the social value of the resource. Research by the Bloomington school, however, demonstrates that the commons is not, by default, tragic, but instead presents an opportunity. McGinnis (1999) collects the many successful common fisheries, water basins, and other shared resources that the Bloomington school has examined and analyzed over the past decades. Their lengthy studies have culminated in constructing a grammar of institutions, the Institutional Analysis and Development (IAD) framework Hess and Ostrom (2007a).

Ostrom (1990, 90), Hess and Ostrom (2007b, 7), and Wilson et al. (2013, S22) summarize eight "design principles" for institutions to successfully manage common pool resources. (Dourado and Tabarrok, 2014, 19) call these "the eightfold path to success." Long-enduring institutions that successfully manage common pool resources have the following features:

1. Clearly defined boundaries for group membership and for the shared resource
2. Proportional equivalence between benefits/costs and the contributions/transgressions of members
3. Collective-choice arrangements to allow members to establish rules and make decisions for the group
4. Monitoring of member behavior through detectable norm-abidement to prevent free-riding
5. Graduated sanctions for transgressors ranging from informal gossip to expulsion
6. Conflict-resolution mechanisms that are viewed as efficient and fair
7. Recognition of rights of group members to self-organize internally
8. Polycentric relations between the group and other social orders to maintain optimal size and autonomy

In the next section, I investigate the success of Wikipedia by applying each of these principles to the rules, norms, and and history of Wikipedia.

3 Institutional Entrepreneurship: The Case of Wikipedia

Wikipedia, "the free encyclopedia" stands as a testament to entrepreneurship that persists in spite of the risk of an anti-commons from copyright and the traditional theory that precludes its success. Rimmer (2009) argues that Wikipedia emerged in an adverse legal environment and its success has recommends new policy prescriptions that better facilitate the success of collective authorship.

While many serious scholars are beginning to liberally refer others to Wikipedia articles as good primers for intellectual topics, scholarship on Wikipedia itself is widely dispersed and idiosyncratic. Among economists, Wikipedia itself is often relegated as an interesting example used briefly in the service of some larger point, whether for modelling contribution spillovers (Aaltonen and Seiler, 2014), estimating the relationship between

group size and contributions (Zhang and Zhu, 2011), expanding the theory of the firm (Frey et al., 2011), examining club competition and individual recognition (Polborn, 2007), applying network theory (Ransbotham et al., 2012), or discussing the role of user generated content in the rise of the Web 2.0 (Kaletka and Pelka, 2011).[9] Others have primarily focused on empirically assessing whether Wikipedia is factually accurate and unbiased, relative to other encyclopedias. Giles (2005), writing for the prestigious *Nature* found that, on average, the difference in the accuracy of 42 science articles on Wikipedia and their counterparts in the prestigious *Encyclopedia Britannica* "was not particularly great." Greenstein and Zhu (2012), writing for the *American Economic Review* found that on net, wikipedia remains neutral, interestingly *not* because of the revision mechanism within individual controversial articles, but because for every biased article on an issue, there appear equal and opposite biased articles on that issue. However, Hasty et al. (2014, 371) caution that "[m]ost Wikipedia articles for the 10 costliest [medical] conditions in the United States contain errors compared with standard peer-reviewed sources." Only a handful of economists or lawyers have briefly mentioned Wikipedia's impact on intellectual property rights, again often as a utilitarian example (see e.g. Benkler (2006); Tapscott and Williams (2007); Shirky (2008); Frey et al. (2011); Polborn (2007); Boldrin and Levine (2008); Dourado and Tabarrok (2014)).

The three main components that define Wikipedia are its use of the "wiki," its self-conscious efforts to be a wikipedia, and the licensing strategy for its content (Benkler, 2006, 70-71). A wiki, named after the Hawaiian word for "quick," is a web application created by Ward Cunningham that allows multiple users to collaborate on a document. The wiki neither structures the organization of the document, nor allocates ownership rights to any individual or entity. Any user with access can write, modify, and delete portions of the document anonymously (though typically they must register under a username or pseudonym). Each document is bundled with a "talk" page, where users can discuss the actions and edits of other users. All actions and changes to the document by users are publicly recorded and archived, allowing any user monitoring changes to quickly revert the document to previous states if necessary (for instance, to reverse blatant vandalism). Figures 6-8 demonstrates the features of the wiki application via the article on "Economics"[10] on the English Wikipedia.

[9] In fact, there is a Wikipedia article on academic studies of Wikipedia: `http://en.wikipedia.org/wiki/Wikipedia:Academic_studies_of_Wikipedia`

[10] `http://en.wikipedia.org/wiki/economics`

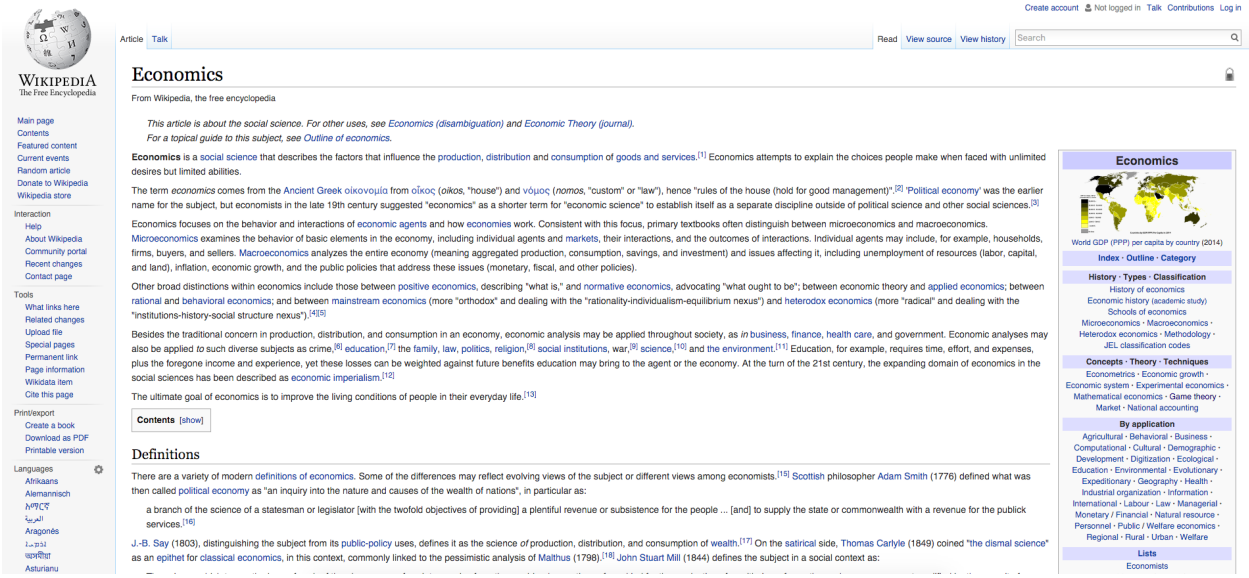


Figure 2: Screenshot of main article page for example article on “Economics” as of November 1, 2015. Note the “Talk”, “Edit”, and “View History” options on the top menu bar.

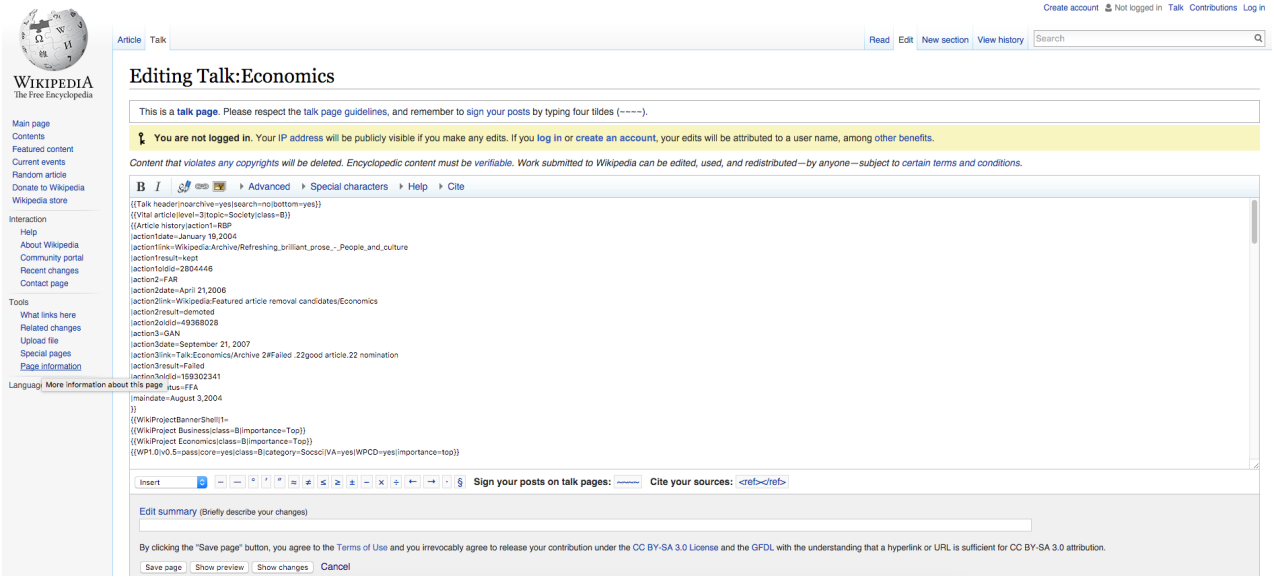


Figure 3: Screenshot of edit page for example “Economics” article. Any user (note I am not even logged in) can edit the article page using simple Markdown language.

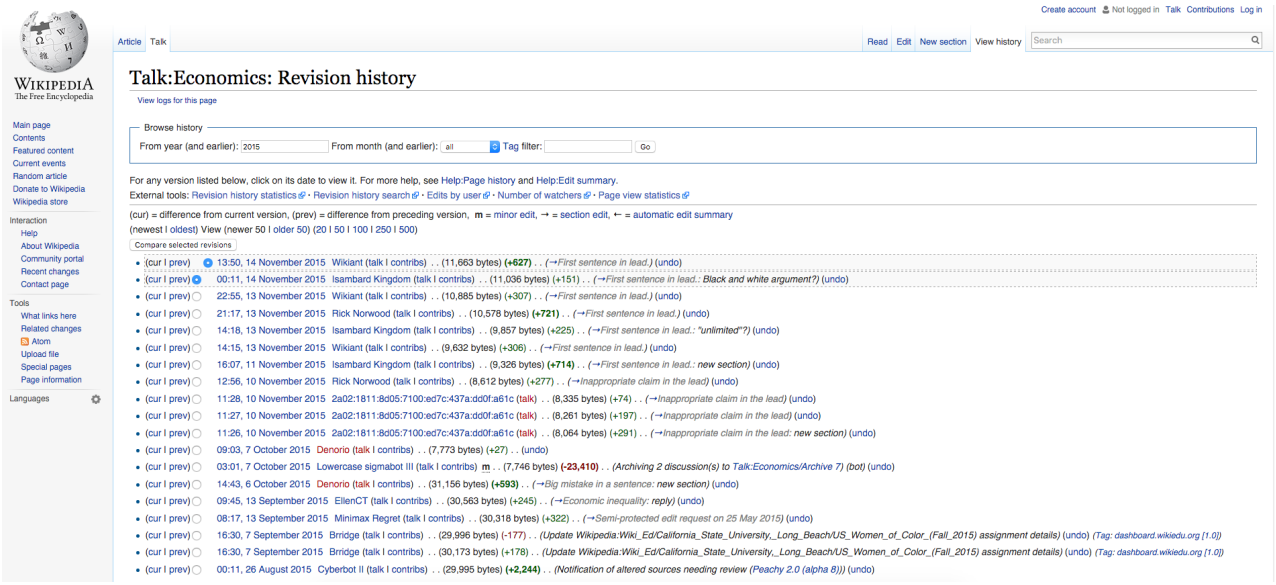


Figure 4: Screenshot of history page for example "Economics" article. Any user can observe the edits by all users by username and IP address, as well as reverse them

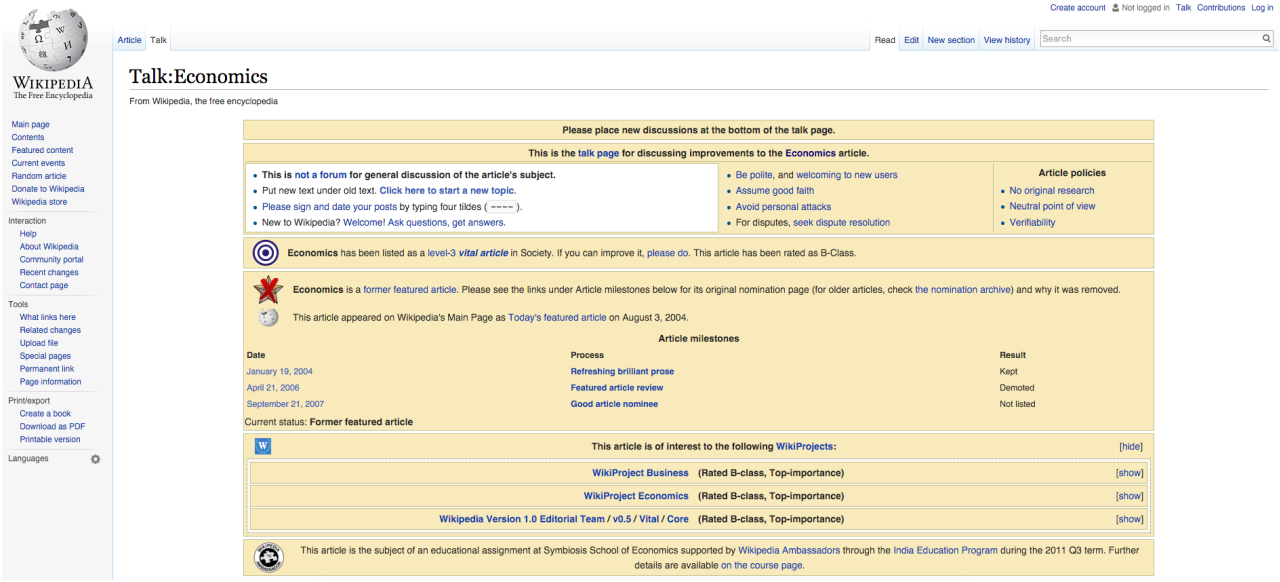


Figure 5: Screenshot of talk page for example "Economics" article. Editors discuss the changes everyone is making to the article page. Note that this screenshot simply covers some of the table of contents for a long talk page

The second key feature of Wikipedia is the deliberate effort by participants with a unifying collective goal: to disseminate knowledge. This generates a community of interested members, which enables co-production by users to generate the content of the encyclopedia articles.

Finally, all text on Wikipedia, by virtue of being "fixed in a tangible medium of expression" (17 USC §102) is automatically copyrighted under U.S. law (and the Berne Convention internationally). However, the wiki software does not establish a sole owner that can control the work, "no editor owns an article and any contributions can and will be mercilessly edited and redistributed."[11] Furthermore, Wikipedia employs a compulsory license, the Creative Commons Attribution-ShareAlike 3.0 Unported License (CC BY-SA).[12] This license itself is a private entrepreneurial creation by Lawrence Lessig, Hal Abelson, and Eric Eldred, who collectively founded the Creative Commons, a nonprofit organization that provides licenses which promote greater access and distribution of copyrighted works. The CC-BY-SA license permits users to share and adapt the work without requesting permission, even for commercial purposes, so long as they attribute credit to the source and document any changes. Furthermore, derivative works based on the original must also employ the same license conditions as the original. This ensures that all Wikipedia articles remain open to satisfy the goal of becoming a freely accessible public encyclopedia, and its content can never be closed off into an anti-commons by users or copyright holders.

Wikipedia is governed by its members adherence to the "five pillars" that normalize articles and interactions on the site:[13]

1. Wikipedia is an encyclopedia
2. Wikipedia is written from a neutral point of view
3. Wikipedia is free content that anyone can use, edit, and distribute
4. Editors should treat each other with respect and civility
5. Wikipedia has no firm rules

I now turn to examining how Wikipedia's success in terms of how it (intentionally or otherwise) implements all eight of the Bloomington School's principles for successful institutions.

3.1 Defined Boundaries

The definition and purpose of Wikipedia is made quite clear by its first pillar – it is a voluntary encyclopedia open to all to use and edit. In further defining the purpose of building an open online encyclopedia, Wikipedia

[11] http://en.wikipedia.org/wiki/Wikipedia:Copyrights
[12] https://creativecommons.org/licenses/by-sa/3.0/us/
[13] http://en.wikipedia.org/wiki/Wikipedia:Five_pillars

has a made clear list of what it is *not* meant to be.[14] Wikipedia is not, for instance, a dictionary, a place to public original research, a soapbox, a newspaper, a manual, or a technical journal.

Wikipedia exhibits perhaps one of the clearest possible examples of coproduction. Coproduction allows consumers or end-users of a commons to input skills or information into the production of public goods (Aligica and Tarko, 2013). For many public goods, such as Wikipedia articles, the input of consumers is essential "if there [is] to be any production at all" (Parks et al., 1981, 1001-1002). This is highlighted precisely by the fact that Wikipedia deliberately refers to all users who use Wikipedia as *editors* – a title that outside of Wikipedia connotes gate-keeping authority and discretionary power. Indeed, without this key element of blurring the line between consumer and producer, Wikipedia would have gone the way of Nupedia long ago, and languished in the shadow of authoritative expert encyclopedias like the *Britannica*.

Wikipedia is also a shared resource that is potentially available to any user to consume. Any person who visits the website is automatically a user of Wikipedia, and catalogued according to their computer's IP address. They can consume Wikipedia's content anonymously. In principle, any of these users can also contribute and edit articles. At its original inception, Wikipedia was truly "anarchic" in that anyone could create and edit articles without any formal registration requirement. However, over time, Wikipedia instituted a policy requiring users to register a username (usually pseudonymous) with the website in order to create new articles, edit certain articles, and acquire other privileges.[15] The site has also instituted further exceptions to open editing, as in cases of suspected vandalism, described below.

A 2010 voluntary survey of 170,000 self-selected Wikipedians across 231 countries and in 21 languages provides some of the first comprehensive demographic information of who uses and who contributes to Wikipedia (Glott et al., 2010). Most users (about 65%) identified themselves as only a reader, with 23% reported themselves as occasional contributors, and a small minority 7% called themselves regular contributors (5). Readers are on average 25 years old, with about half of respondents younger than 22 years (7). Readers tend to be slightly younger (on average 25 years old) than contributors (on average 26 years old). There is also a disproportionate percentage of male readers (69%) and contributors (86%) relative to female readers (30%) and contributors (13%), as depicted in Figure 6. Furthermore, contributors are slightly better educated than readers, with most users having at least a high school or Bachelor's level of education, reflective of the dominance of younger age groups, as shown in Figure 7. Additionally, while users report a variety of unique reasons they choose to contribute, most resemble desires to share and spread knowledge and fix errors, while very few contribute for the purpose of earning money, furthering a career, or, interestingly, even to gain reputation among the Wikipedia community (9). For those who do not contribute, the most common

[14] http://en.wikipedia.org/wiki/Wikipedia:What_Wikipedia_is_not
[15] http://en.wikipedia.org/wiki/Wikipedia:Tutorial/Registration

reasons were unfamiliarity with the process, lack of time, uneasiness about editing others work, the beliefs that they are not qualified or do not know enough about a subject to contribute, and the satisfaction with simply reading rather than getting involved (10). Those users that do contribute edits tend to make only a handful of them, with a very small proportion of users making many contributions, as shown by Figure 8. The left panel shows the proportions of of users who contribute a certain number of edits and the right panel shows the proportion of total edits made by those users who contribute a certain number of edits.[16]

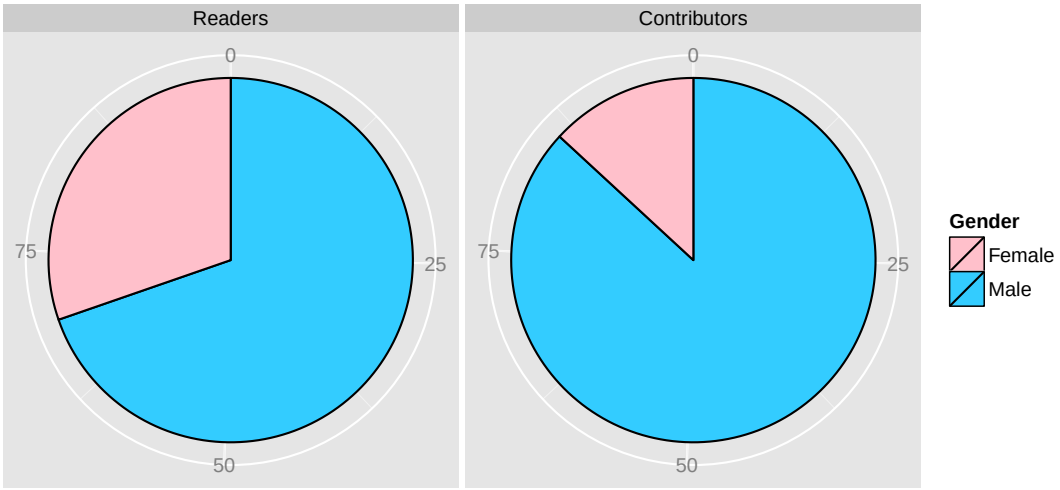


Figure 6: Breakdown of Wikipedians by gender, compiled from Glott et al. (2010, p.7)

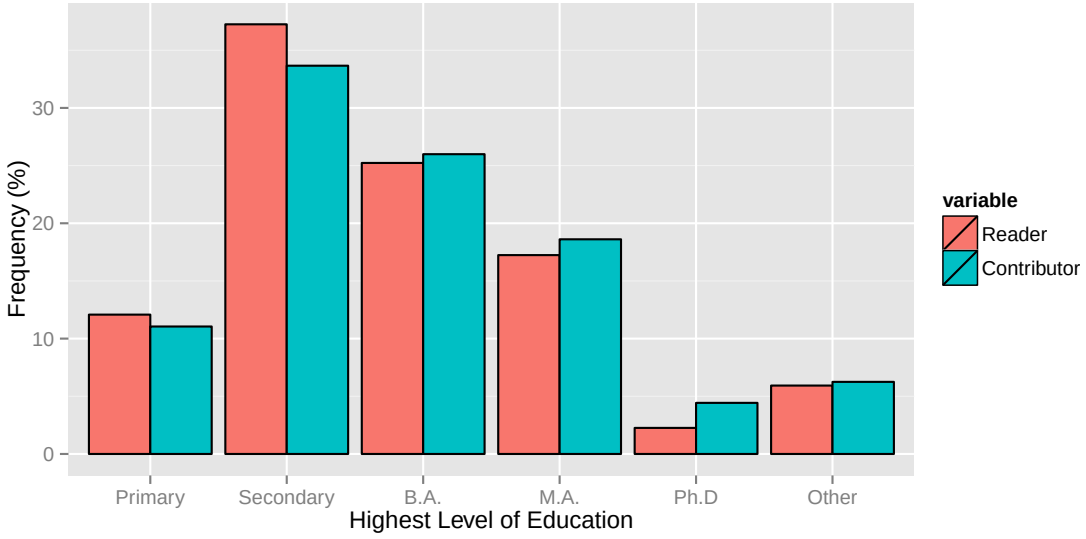


Figure 7: Breakdown of Wikipedians by highest education attained, compiled from Glott et al. (2010, p.7)

[16]Data compiled from `http://stats.wikimedia.org/EN/TablesWikipediaEN.htm`

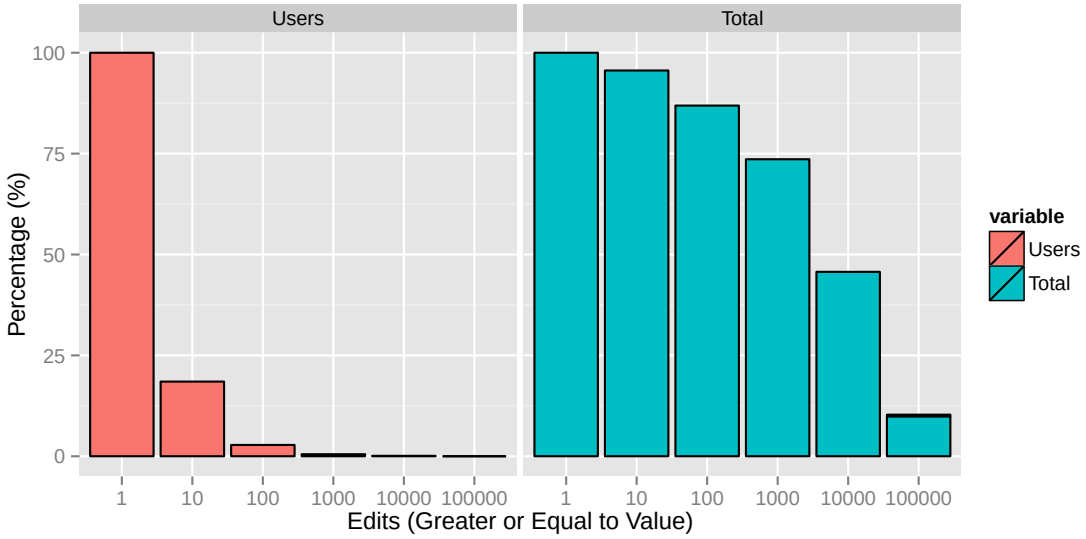


Figure 8: Distribution of article edits by registered non-bot Wikipedia users. The left panel shows the proportions of of users who contribute a certain number of edits and the right panel shows the proportion of total edits made by those users who contribute a certain number of edits.

Stemming from Wikipedia's intentional collective goal of being an encyclopedia, Wikipedians will use several mechanisms to exclude users and editors who run contrary to this goal. Primarily, users who deliberately and systematically try to edit Wikipedia articles for any reason other than a good faith addition of accurate information (i.e. spreading lies, rumors, biased views, edits-for-hire, etc), will have their IP address and (if applicable) username banned - which will allow them to read but no longer edit articles. This process is described in more detail in the next section.

3.2 Proportional Costs & Benefits

Proportional equivalence between costs and benefits ensures that certain subsets of the group are not being bearing disproportionate costs or benefits, which would threaten the internal stability of the group and its shared resource.

As David (Hume, 1740, Book III, Part II, §V) famously described,

> "Men being naturally selfish, or endowed only with a confined generosity, they are not easily induced to perform any action for the interest of strangers, except with a view to some reciprocal advantage, which they had no hope of obtaining but by such a performance."

People thus reserve their benevolence for those closest to them, and can often foster cooperation with others in the broader commercial society according to anonymous arms-length exchanges in markets according to prices (Hayek, 1973).

Wikipedians do not get paid, even the Wikimedia Foundation, which owns the physical infrastructure behind Wikipedia and oversees its operation, is a non-profit organization financed by donations. Thus, Wikipedians do not contribute edits motivated by financial interest. Scholars have long observed that in many cases of expressive works, such as analogously writing a book, authors do not necessarily require financial motive to produce such a public good (see e.g. Landes and Posner (1989, 41-50) and Johnson (2012)). Wikipedia's structure seems to have tapped the right ingredients to captivate Hume's "confined generosity" to motivate individuals to contribute in the public interest.[17] Primarily, this is because Wikipedia is modular and additive.[18] Unlike collecting more burdensome financial contributions to fund a public bridge, Wikipedia is able to channel this confined generosity to an extent due to the small modularity of any marginal contribution by a user. An article on Wikipedia is an independent good, and is capable of being effectively produced by one person or one million. Furthermore, each individual by any given user contributes to the social product (provided that edit is not intentionally malicious), and an individual edit can take as little or as much time as a reader has time (and passion for). This could range from a 30 second correction of a misspelled word, to hours-long creating of a new article that the user is passionate about. Researchers studying free and open source software have long noted that modularity is a key component of successful non-proprietary software (Raymond, 1999; Benkler, 2002). Thus, the ease, modularity, and additive nature of editing Wikipedia helps motivate users to edit for the social good out of their own personal desires without requiring financial incentives (which would be necessary if making contributions were of a larger and individually costlier scale).[19]

The main currency within Wikipedia's community is reputation and social standing within the community, for a small subset of passionate and active users. Building a reputation, conceived as investing in social capital, can allow an editor to engage in a lot of activities that other users can not. First, she can expect a higher likelihood of her edits to be approved and not removed. Additionally, she can "run" for higher office as a bureaucrat or an administrator. This, however, can only account for a small fraction of a percent of all registered users (not even counting unregistered readers). As self-reported by users in the survey by by the reasons given for contribution/noncontribution in Glott et al. (2010) above, few users chose to contribute out of a desire for reputation within the Wikipedia community. Hence, any potential gains to reputation are

[17] Among academics, anonymous peer review is a highly visible demonstration of how salient this non-financial incentive-structure can be for highly-motivated people. Although academics do not get paid for reviewing academic work (except in rare cases), there is a strong personal motivation to critique and improve existing work for the love of learning and disseminating truth. Often, much like contributing a snippet to a Wikipedia entry, an academic review requires only to read and critique one paper, a sufficiently minor task to not require excessive thanks or payment. I thank one anonymous reviewer for pointing this out, and two for engaging in such an activity.

[18] I thank Alex Tabarrok for emphasizing this key distinction between Wikipedia and other public goods.

[19] The meteoric rise of social media also points to the salience of such motivational structures. Intense political arguments on Facebook and Twitter frequently arise and even have become cited in journalistic outlets as genuine news, due to their low time cost of participation and high reputational (non-monetary) and "warm glow" rewards.

only valuable to those who wish to be very active, visible, and well-ranked among the Wikipedia community – precisely those interested in admin powers. Most users are satisfied with simply get by with reading or editing to feel good, rather than seeking to join a close-knit community (which one can participate in only by actively posting and discussing on talk pages).

Wikipedia has a small degree of hierarchy that naturally emerges from within the community, in the form of functionally-defined roles of administrators ("admins") and bureaucrats. Admins are users that are granted greater control over articles and other user abilities, such as the power to block and unblock user accounts and IP addresses from editing articles, protect and unprotect pages from editing entirely, delete and undelete articles, and rename articles.[20] At Wikipedia's inception, every user was granted admin powers, but over time, as a protection against vandalism, these powers have been trusted to only a small number of dedicated users. Wikipedia does not have any formal rule for qualifying a user for admin powers, every user can apply to become an admin, but the community holds a discussion about each application and if there is a clear positive consensus after 7 days, that user is approved. Thus, only members who are deeply involved in the community and make frequent contributions are likely to be rewarded with adminship. Jimmy Wales argues that the administrator is merely a technical role played by regular community users and is reserved for editing emergencies, rather than any inherent authority or hierarchical superiority. The role, like the rest of Wikipedia, is purely voluntary and uncompensated. The English wikipedia currently has 1,463 administrators.[21]

Bureaucrats are users simply invested with the power to add and remove administrator powers to a user to enact a community consensus.[22] The process of becoming a bureaucrat is similar to that of requesting adminship, though a higher degree of community consensus is required for this role. There are currently 33 bureaucrats on the English wikipedia.

3.3 Collective-choice Arrangements

All successful groups need some sort of decisionmaking procedure to create and modify rules that affect all members of the group in furtherance of the group's ends. The main way Wikipedia's ends are achieved is through informal community consensus on article content and user activities.[23] As there are few formal rules that govern the editing process (mostly reserved for resolving crises), most rulemaking is emergent and norm-based, contingent on the overall goal of creating an open encyclopedia.

Following the tradeoff between internal decisionmaking costs and external costs identified by Buchanan

[20] http://en.wikipedia.org/wiki/Wikipedia:Administrators

[21] Some fear that the declining number of admins is worrisome, see e.g. http://www.theatlantic.com/technology/archive/2012/07/3-charts-that-show-how-wikipedia-is-running-out-of-admins/259829/

[22] http://en.wikipedia.org/wiki/Wikipedia:Bureaucrats

[23] http://en.wikipedia.org/wiki/Wikipedia:Consensus

and Tullock (1962), Wikipedia features quite low internal decisionmaking costs and high external costs.[24] As typified by Wikipedia's first pillar, Wikipedia's collective goal of being a genuine high-quality encyclopedia places hard constraints on articles and editors: edits that are commonly viewed as detracting from this goal are fiercely resisted and will be removed by nearly anyone. Thus, external costs are rather high. Despite this, internal decisionmaking costs are quite low, since the number of people interested in any given page is quite small (certainly this group size varies across article subjects from "Tholeiitic magma series" to "Capitalism"). Additionally, Wikipedia is able to achieve its goal of a neutral point of view by including *all* perspectives that heterogeneous users are willing to contribute into an article. Often, any controversial article ("Capitalism", for instance) will feature a "Criticism" section where users of different opinions can choose to express their dissent through opposing arguments. As a result, with relatively high external costs and low decision costs, the *Calculus of Consent* model predicts that the optimal decisionmaking rule is to require agreement among a very large proportion of the deciding group. Wikipedia confirms this prediction, as it is a foundational feature of Wikipedia that major decisions are made "by consensus."

Wikipedia delineates several tiers of a normative structure to govern user interactions: policies, guidelines, and essays.[25] Policies "have wide acceptance among editors and describe standards that all users should normally follow." Guidelines are "sets of best practices that are supported by consensus," that users are encouraged to follow, but common sense is mostly needed. Essays are merely opinions of a single user or group of users about an issue that has not reached community consensus. Users can post essays that they do not wish others to edit on their own personal user page.

Even at a higher, constitutional level, to change overall policy, Wikipedia does not employ any sort of formal voting system. Wikipedians can propose and discuss changes to overall Wikipedia policy or guidelines on relevant talk pages. "Village pump" pages exist as a forum for members to provide notice and discussion of proposals for major rule changes.[26]

3.4 Monitoring

The success of any group effort strongly depends upon the ability of members to cheaply monitor defection and behavior harmful to the collective goals of the enterprise. One might reasonably imagine that Wikipedia faces a gargantuan collective action problem in monitoring the accuracy of millions of articles that are subject to hostile edits by anonymous readers at all hours of the day. However, several technical components of the wiki application software mitigate this problem, as well as the different incentives of individual users and groups.

[24] I thank Vlad Tarko for emphasizing the predictive power of the Calculus of Consent model for this case.
[25] https://en.wikipedia.org/wiki/Wikipedia:Policies_and_guidelines
[26] https://en.wikipedia.org/wiki/Wikipedia:Centralized_discussion

Monitoring is still important for Wikipedia, despite the fact that neither its product nor its members' contributions are denominated in measurable dollar terms.[27] In meeting their collective goal of being a genuine encyclopedia, Wikipedians must still monitor the quality of each other's articles. Fortunately, the wiki platform enables the site to distribute monitoring capacities across its members with relevant knowledge and interests.

As part of the wiki platform, each article has a talk page, where users can comment on changes by other users, as well as view an archive of all changes made by ever user according to their username or IP address (if unregistered). Users can also add a particular article to their "watchlist," where they will automatically be notified of any change to the article's content, and of conversations occurring on the article's associated talk page. Edits to an article that are malicious can easily be detected and reversed with the click of a button, reverting the article to its state before the particular edit. Figures 6-8 again illustrate this with an example.

One might ordinarily expect a strong collective action problem to arise between a single committed vandal and a large group of people benefitting from a common resource such as a Wikipedia article. With many free riding readers of any given article, it might be a potential challenge for users to police and combat malicious changes in any given article. However, the design of the wiki software flips the logic of collective action problems on its head, giving users an effective means to quickly reverse vandalism through their watchlists of articles they have an interest in, with this ability increasing with reader size.

Though all users have editing power, there are no editors that are tasked with overseeing the entire set of articles. Typically, any given contributor is interested only in a particular set of topics that they have expertise, interest, or passions in. Should a malicious user attempt to vandalize and intentionally edit an article for the purpose of mockery, irrationality, or prejudice, these edits can be quickly discovered by interested users. The larger the audience of interested readers on generic, popular, or controversial topics (e.g. capitalism, Star Wars, or current events), the more eyes policing and the less likely a vandalistic edit can seep through the cracks. More technical, idiosyncratic, and less trafficked articles face a higher likelihood of vandalistic edits persisting for a significant period of time. Furthermore, users that repeatedly vandalize articles are identified by their username or IP address, which can be suspended or permanently banned by admins (see sections 3.5 and 3.6).

A second type of monitoring dilemma exists between ordinary users and admins. Ordinary users must have some recourse over admins that have gone rogue or are otherwise abusing their admin powers, such as improperly banning users or excessively locking articles. Admins are monitored by other users who are interested in respective articles that are being abused, including other admins. Admins suspected of continual

[27] I thank Joy Buchanan for raising this issue.

abuse of their powers can be removed by Jimmy Wales or the arbitration committee (see below).

3.5 Graduated Sanctions

In order to ensure compliance with community norms, any group must have punishments to deter unsocial behavior. Punishments for violations further must be scaled gradually, to ensure they do not further incentivize worse behavior (e.g. minor infractions should not be met with immediate expulsion, or else there is no relative cost difference between committing a minor or a major infraction).

Wikipedia has three main sanctions against individual violators: communal shaming, temporary bans, and permanent bans. Individual users that vandalize articles or otherwise violate Wikipedia's policies are first often met with discussion on the article's talk page. A user's malicious edits will typically be reversed by another user and documented on the talk page. For continual vandalism by a particular user, identified either by their username or IP address, admins can block the username/IP address from editing temporarily, or for severe cases, permanently. Wikipedia publicly posts all active sanctions in place against violators.[28]

3.6 Conflict-resolution Mechanisms

Any group must have mechanisms to resolve disputes between individual members that are effective at resolving conflict, and are viewed by all participants as impartial and decisive. In general, with respect to its mission of creating an open source encyclopedia, Wikipedia strives to minimize potential conflict over the content of articles by mandating a "neutral point of view."[29] Rather than aiming for consensus on a single point of view and subsuming all content to it, Wikipedia encourages a diversity of opinion. To productively channel the passions of competing factions of opinion, Wikipedia articles often feature sections that include the various arguments for different opinions.

The main type of conflict that naturally and predictably emerges is a disagreement over the content of an article. This can range from honest disagreements between varying perspectives on an issue (e.g. Republican and Democrat users editing an article on U.S. politics) to deliberate "trolling" and vandalism of an article. To the extent that different opinions are not productively expressed in designated criticism sections of an article, users attempt to resolve it in the talk page discussion. As noted above, when this does not work, it falls to admins mete out punishment of uncooperative users.

As both monitoring and control over content are distributed across potentially all users, conflicts over article content can easily trigger an "edit war," where multiple users each repeatedly edit articles in their

[28] https://en.wikipedia.org/wiki/Wikipedia:Arbitration/Active_sanctions
[29] https://en.wikipedia.org/wiki/Wikipedia:Neutral_point_of_view

favor and reverse all edits from other users expressing dissident content.[30] This includes cases where an individual editor repeatedly restores his preferred version of the article after it has been removed by other editors. Wikipedia normally encourages editors to come to a consensus on the talk page and essentially "talk-it-out." When disagreements turn into outright edit wars, Wikipedia employs a "three-revert rule" (3RR), a bright line rule which declares that no editor may perform three reversions of others' content on one article in a 24 hour period.[31] Editors who frequently edit war and/or violate the 3RR rule are subject to the graduated sanctions listed above.

Additionally, for articles on topics, persons, or events that systematically cause controversy (the reader can certainly imagine them), articles can be placed under "protection."[32] While any user can call for an article's protection in its associated talk page, it is up to an admin to change the protection settings for an article. Protection constitutes a range of measures, ranging from preventing any changes by unregistered users without admin approval to full protection where only admins can edit articles.

Wikipedia's "court of last resort," at least on the English Wikipedia, is the Arbitration Committee (ArbCom).[33] The committee is comprised of 12 users appointed by Jimmy Wales after advisory elections by the community, and to date has decided over 370 cases of user conduct. The committee's decisions are binding on Wikipedia members, and their punishments have ranged from warnings to bans. Hoffman and Mehra (2010) document that the arbitration committee largely stays true to its ideal by choosing to review primarly behavioral disputes between user conduct, rather than the content of articles under controversy.

Wales in particular has quite a lot of *de jure* control over Wikipedia. As he appoints the Arbitration Committee members, although constrained by selecting those members that have been approved by community consensus, he could potentially wield the ArbCom's power to ban any users that post content he disagrees with. However, use of the ArbCom has been extremely limited (although not without some critics).

Wales appears to act as a benevolent despot, although he has rejected commentators' use of this term to describe him, and instead chooses to compare himself to the Queen of England, a constitutional monarch with no real power.[34] If we can assume, based on his statements and actions, that Wales' objective function is to maximize the spread of information,[35] then the literature on rational dictatorship implies that he will select the optimal amount of interference with production (see e.g. Olson (1993); Tullock (2005); Wintrobe (1990); Congleton (2011)). Unlike a dictator over a country, however, Wales is not in the strict sense a residual claimant to the financial success of Wikipedia, as Wikipedia is a non-profit organization that produces no

[30] http://en.wikipedia.org/wiki/Wikipedia:Edit_warring
[31] http://en.wikipedia.org/wiki/Wikipedia:Edit_warring#The_three-revert_rule
[32] https://en.wikipedia.org/wiki/Wikipedia:Protection_policy
[33] https://en.wikipedia.org/wiki/Arbitration_Committee
[34] See http://slashdot.org/story/04/07/28/1351230/wikipedia-founder-jimmy-wales-responds
[35] http://en.wikipedia.org/wiki/Jimmy_Wales

revenue-generating assets.[36] Wales thus would choose to engage in minimal intervention into the production of Wikipedia articles – only enough to ensure basic order and prevention of severe vandalism that would jeopardize information creation and spread. Furthermore, to ensure that this is a credible commitment on his part, Wales empowers the ArbCom to veto his actions and have members of the ArbCom selected by popular vote, similar to the evolution of the constitutional King and Council models in Congleton (2011). This ensures that editors are convinced that Wales will credibly restrain his authority and will all continue to produce articles, increasing the likelihood of achieving Wikipedia's mission, and satisfying Wales' preference for disseminating information (not to mention, presumably filling the coffers with donations).

3.7 Recognition of Self-Organization Rights

All successful groups must have the authority to conduct their own affairs and their members autonomy respected. As stated in the fifth pillar, Wikipedia does not have firm rules, but instead has mere "policies and guidelines...not carved in stone." Contingent on respecting the basic rules of common sense and human decency, with wanton exceptions listed above, the openness of Wikipedia achieves respect for individual autonomy in spades, with its categorical rule "to ignore all rules."[37]

As noted above, the English Wikipedia's Arbitration Committee has only decided 370 cases of user conduct, despite the existence of millions of articles, tens of millions of users, and almost two billion edits. This indicates that Jimmy Wales and the Wikipedia project do indeed respect users rights to self-organize in practice, not just in theory or PR statements.

3.8 Polycentric External Relations

For those groups that interact with other groups and with the social order at large, the given group must have mechanisms that coordinate activity productively across different groups such that the size of various groups are proportional to the size of the collective problem faced. Wikipedia is an internet website that produces content available to all with an internet connection across the world. As the raison d'etre of Wikipedia is producing expressive content in the form of original encyclopedia articles, Wikipedia must interact with other media outlets and original works under national and international copyright laws. Since all original expressive works require inputs from previous existing works, Wikipedia users and article content must not run afoul of existing expressive works or their copyright holders.

Creators often view copyright as the sine qua non of creative work, as having a copyright on a work

[36] Wales does, however, certainly benefit from donations to the Wikimedia Foundation and in reputation gains from Wikipedia's success.

[37] `http://en.wikipedia.org/wiki/Wikipedia:Ignore_all_rules`

stands as evidence of expressing something of value.[38] On Wikipedia, however, this is far from the case. While there are no exclusion rights over Wikipedia articles, and there are no monetary exchanges evidencing value, it is almost self-evident that Wikipedians are creating value in the form of zero-price encyclopedia articles. While their individual contributions to various articles may be anonymous and unknown to the outside world, users continue to provide valuable content. Within the Wikipedia community, some of these contributions are recognized and allow particularly prominent editors to rise to the ranks of administrators and bureaucrats. Perhaps due to the modularity of any marginal Wikipedia edit, users can create value and contribute to the Wikipedia project on their own time solely due to their "confined generosity" to use Hume's term, and their desire to further the expansion of knowledge. Were they asked, or *ordered*–as in an employment relationship–to write substantial portions of encyclopedia articles, it seems more likely that contributors would feel the stronger burden of their opportunity cost, and require a financial incentive.

In any event, Wikipedia does however make use of other media and expressions created outside its website, which are governed by the law of copyright. Wikipedia policy requires users seek permission prior to posting any copyrighted work (text, image, video, or sound recording) within a Wikipedia article.[39] Wikipedia strongly encourages its users to use works that are in the public domain, works that can be plausibly used under copyright's "fair use" doctrine, works that users themselves have ownership over and authorize for use on Wikipedia, or copyrighted works that are licensed for free use (such as works under Creative Commons licenses). The license also mandates that authors of content on Wikipedia must adhere to Wikipedia's license, preventing them from closing down the article or seeking royalties from use in an article. Owners of existing works that are posted on Wikipedia without consent have recourse to a special request to take down a Wikipedia article for copyright infringement.

Finally, as mentioned above, Wikipedia content is protected by the Creative Commons CC-BY-SA license. Articles that are produced on Wikipedia are freely accessible to anyone to use, distribute, and modify, so long as they give attribution to Wikipedia and all modifications follow these conditions as well.

4 Implications

The language of "intellectual property rights," makes it seem natural to extend the logic of property rights to the ethereal realm of non-rivalrous ideas and expressions. The common parlance entices many economists, conditioned by the association of "property rights, good; no property rights, bad," to support intellectual property rights. Economists have long recognized the critical role that property rights have played in both the successes and failures of attempts to cultivate economic growth (North, 1990; Acemoglu and Johnson, 2005).

[38] I thank one anonymous reviewer for pointing our the significance of this point.
[39] `https://en.wikipedia.org/wiki/Wikipedia:Copyrights`

Economic resources, that have yet to be "propertized" remain in the commons, subject to the "tragedy of the commons," where each user faces a personal incentive to extract as much of the resource as possible before others do likewise as no user can exclude another. Hence, the resource gets depleted.

Extensive research by the Bloomington School, however, demonstrates how informal institutions and entrepreneurship over the rules of the commons presents an "opportunity," not necessarily a tragedy. If these rules are improperly designed, or imposed from on-high without regard to the facts on the ground, then the tragedy of the commons – where weak rules under-exclude, or a tragedy of the anti-commons, where strong rules over-exclude. Thus, the key to understanding economic growth and the role of property rights is not to ask the simple binary question "do property rights *exist* for X?" or exist at some theoretical optimum, but how those rights *fit* within an existing framework of institutions that promote or stifle cooperation, exchange, and production (Buchanan, 1964; Boettke et al., 2004). In doing so, scholars in this tradition have sought to study many cases of both market and institutional entrepreneurs and communities designing "good" rules of interaction that promote cooperation, exchange, and proper stewardship over common resources (see e.g. McGinnis (1999); Frischmann et al. (2014); Dourado and Tabarrok (2014)).

Wikipedia should stand out to the modern reader like a sore thumb as one of the quintessential examples of institutional success in the knowledge economy of the early 21st century. It is a resource that is used by millions at a money price of zero, and yet creates unquantifiable but significant value in spreading knowledge to academics and laymen alike. Its success stands as an existence proof that the traditional economic model of intellectual property, where creators will not create unless they are legally permitted to exclude others from using their product, has important limitations as a descriptive theory, and potentially even as a normative one.

Copyright has both made it more difficult for entrepreneurs to navigate the market process due to higher transactions costs of negotiating rights, as well as distorting the channels of activity by major creative firms and entrepreneurs (Safner, 2015). In the midst of rapid technological change and potential entrants, many established studios and production houses have found it a far more profitable venture to continue extracting rents through the political and legal process of enforcing existing copyright claims than by out-innovating competitors in the marketplace (Boldrin and Levine, 2008). Their activity has led dangerously close to a tragedy of the anti-commons (Heller, 2008) of incumbent rightsholders placing future innovation at risk due to the influence of their "lobbynomics" (Hargreaves, 2011). Despite this, while plenty of incumbent firms continue to cling to business models rapidly being made obsolete by new technology and use the legal-political process for rent-extraction, others have discovered new enterprising ways of circumnavigating this increasing problem. Some have embraced new technologies emerging in the market such as streaming (e.g. Netflix and Spotify) or bundled greater services and convenience to their products (e.g. iTunes and Amazon). Finally,

others have attempted to create new organizational and institutional spaces by redesigning the rules to foster greater innovation and openness.

It is this final form of entrepreneurship, institutional or"higher-tiered" entrepreneurship that this paper has attempted to highlight with the case of Wikipedia. This form of creative solution to a complex puzzle has been understudied by scholars of innovation and culture. Despite the fact that Wikipedia still resides within a legal system of copyright, its content is not created with the intent to exclude others from use, quite the contrary. While Wikipedia should certainly not be taken as a model for all production of expressive works, it opens up the possibility of other forms of organization to provide these peculiar goods. The rise of alternative methods of production, such as free and open source software from Linux to LaTeX, in which this article was written, further offer examples of leveraging the traditional framework of copyright to serve ends that its designers had not intended or foreseen.

Hence, we must revise our theoretical frameworks of how expressive goods are produced to account for greater institutional diversity. Closed-form solutions, such as optimizing the tradeoff between incentive and access holding a society's institutional form constant prevents us from greater understanding of cultural expression and its production. Furthermore, we risk destroying additional dimensions of culture-preserving institutions by altering the law along a somewhat two-dimensional model. Instead, we must take a broader, comparative institutional approach to studying efficiency and effectiveness in the provision of expressive works. Following the Bloomington School's theory and methodology, we must continue to examine other case studies to uncover the true breadth of institutional variety and creative potential.

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